
CS246: Database Management Systems Lab

Week # 06 (1 Questions, 78 Marks)

Lab session: AL1

Held on: 16-Feb-2026 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 4

Submission dead line: 17:00 hrs, 16-Feb-2026

Submissions accepted from: 16:45 hrs, 16-Feb-2026

Submissions accepted till: 17:30 hrs, 16-Feb-2026

Every one minute delay in submission will result in one mark deduction

Instructors Dr. V. Vijaya Saradhi

Head TA Divanshu Walia

Department of CSE, IIT Guwahati

This assignment is so designed to have errors. Do not immediately panic and raise questions. Understand what the questions is and apply the theoretical concepts and verify what is being produced aligns with your theory concept or not

Question 1: (78 points)

Data Manipulation Language

Task 01 (1 mark) Create a new database named `week06`.

Task 02 (1 mark) Switch to the database `week06`.

Task 03 (10 marks) Perform the following operations:

1. (3 marks) Create a table `T1_default` with the following specifications

1 st column	fixed length char of size 20	No constraints
2 nd column	integer data type	No constraints
3 rd column	float data type	No constraints

2. (3 marks) Insert the following records

name	roll number	CPI
student_01	270101999	9.8
student_02	270101998	8.9
student_03	270101997	7.6

3. (1 mark) Update the CPI of the `student_03` to 8.1
4. (1 mark) Update the name of the `student_03` to `sudent_03a`
5. (1 mark) Delete record of `student_03a`
6. (1 mark) Insert a record(`student_03`, 270101997, 8.1)

Task 04 (10 marks) Perform the following operations:

1. (3 marks) Create a table `T2_default` with the following specifications

1 st column	fixed length char of size 20	No constraints
2 nd column	integer data type	primary key
3 rd column	float data type	No constraints

2. (3 marks) Insert the following records

name	roll number	CPI
student_01	270101999	9.8
student_02	270101998	8.9
student_03	270101997	7.6

3. (1 mark) Update **student_01** roll number to 270101998
 4. (1 mark) Update **student_01** roll number to NULL
 5. (1 mark) Insert a record (**student_04**, 270101997, 7.1)
 6. (1 mark) Insert a record (NULL, 270101996, 7.6)

Task 05 (10 marks) Perform the following operations:

1. (3 marks) Create a table **T3.default** with the following specifications

1 st column	fixed length char of size 20	not null
2 nd column	integer data type	unique not null
3 rd column	float data type	No constraints

2. (3 marks) Insert the following records

name	roll number	CPI
student_01	270101999	9.8
student_02	270101998	8.9
student_03	270101997	7.6

3. (1 mark) Update **student_01** roll number to 270101998
 4. (1 mark) Update **student_01** roll number to NULL
 5. (1 mark) Insert a record (**student_04**, 270101997, 7.1)
 6. (1 mark) Insert a record (NULL, 270101996, 7.1)

Task 06 (10 marks) Perform the following operations:

1. (1 marks) Create a table **T4.default** identical to the table definition of **T1.default**. You should use **like** construct in table creation. Refer to table creation manual pages.
 2. (3 marks) Insert the following records

name	roll number	CPI
student_01	270101999	9.8
student_02	270101998	8.9
student_03	270101998	8.6
student_04	270101998	8.5
NULL	270101997	7.6

3. (2 marks) Alter table **T4.default** add a constraint that second column (roll number) is a primary key.
 4. (2 marks) Alter table **T4.default** add a constraint that first column (name) is a NOT NULL.
 5. (1 mark) Update **student_01** roll number to 270101998
 6. (1 mark) Update the name of the **student_03** to NULL

Task 07 (18 marks) Perform the following operations:

1. (6 marks) Create a table **ATMs_csv** with **storage engine as CSV** and with the following specifications (2 marks for specifying storage engine)

Column	Data type	Constraint
1 st (Bank)	fixed length char size: 10	NONE
2 nd (lat)	float	NONE
3 rd (long)	float	NONE
4 th (ATM_no)	fixed length char size: 20	NONE

2. (4 marks) Insert the data from the CSV file **ATMs.csv** into the **ATMs** table. To achieve this perform the following
 - (a) Write a C++ program to print several INSERT statements
 - (b) Save all the INSERT statements in a file with extension **load_atms.sql**
 - (c) At the **MySQL** prompt, use the command **source load_atms.sql**;
3. (2 marks) Delete all Bank's with belong to PNB
4. (2 marks) Update all Bank's of SBI whose lat is 17.687194 to 17.677010
5. (2 marks) List the contents of the table **ATMs**.
6. (2 marks) Alter table **ATMs_csv** and place (lat, long) together as primary key constraint.

Task 08 (18 marks) Perform the following operations:

1. (6 marks) Create a table **ATMs_csv** with **storage engine as MEMORY** and with the following specifications (2 marks for specifying storage engine)

Column	Data type	Constraint
1 st (Bank)	fixed length char size: 10	NONE
2 nd (lat)	float	NONE
3 rd (long)	float	NONE
4 th (ATM_no)	fixed length char size: 20	NONE

2. (4 mark) Insert the data from the CSV file **ATMs.csv** into the **ATMs** table. To achieve this perform the following
 - (a) Write a C++ program to print several INSERT statements
 - (b) Save all the INSERT statements in a file with extension **load_atms.sql**
 - (c) At the **MySQL** prompt, use the command **source load_atms.sql**;
3. (2 marks) Alter table **ATMs_csv** and place (lat, long) together as primary key constraint.
4. (2 marks) Delete all Bank's with belong to PNB
5. (2 marks) Update all Bank's of SBI whose lat is 17.687194 to 17.677010
6. (2 marks) List the contents of the table **ATMs**.

Submission Instructions Store all the SQL queries corresponding to all the above tasks in a file named after your roll number with an extension **.sql**. Example, if your roll number is 270101999 then the file name should be **270101999.sql**

You should submit the following:

1. The SQL file containing solutions for all the tasks
2. The SQL file to insert ATM data into ATMs table
3. The C++ file to generate the insert statements
4. Place the above into a zip file named after your roll number. You should submit the .zip file.